

JAYPEE INSTITUTE OF INFORMATION TECHNOLOGY, NOIDA

**MACHINE LEARNING AND NATURAL LANGUAGE PROCESSING
(PROJECT REPORT)**



TOPIC: NLP-Based Emotion and Sentiment Analysis System

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ABSTRACT

This project explores the development of a text-based emotion and sentiment analysis system for depression prediction, leveraging advancements in natural language processing (NLP). It utilizes text extracted from audio inputs to analyze features indicative of depressive tendencies. The system integrates a lightweight DistilBERT model for contextual understanding of text and a BiLSTM architecture for capturing sequential dependencies and emotional nuances. By focusing exclusively on text-based inputs, the system avoids the complexity of multimodal data while ensuring robust and scalable performance in NLP applications. This approach highlights the potential of leveraging text analytics for mental health diagnostics, enabling efficient, accurate, and accessible depression prediction.

INTRODUCTION

The rise in mental health issues, particularly depression, is a pressing global concern, with the World Health Organization identifying it as one of the leading causes of disability worldwide. Early detection of depression can significantly improve outcomes, yet many individuals remain undiagnosed due to barriers such as stigma, lack of access to mental health professionals, and the time-intensive nature of traditional diagnostic methods. These challenges underscore the need for accessible, scalable, and efficient tools to aid in the early identification of depressive tendencies.

Language, a primary medium for expressing emotions and thoughts, serves as a valuable indicator of mental health. Recent advances in natural language processing (NLP) have demonstrated the potential of analyzing text to uncover emotional and psychological states. Unlike traditional multimodal approaches that require a combination of audio, visual, and textual data, focusing solely on text simplifies implementation and expands applicability across a broader range of contexts, such as telemedicine and digital health applications.

This project seeks to leverage the power of NLP to create an emotion and sentiment analysis system capable of predicting depression using text extracted from audio inputs. By combining DistilBERT, a lightweight and efficient transformer-based model, with a BiLSTM network for capturing sequential and emotional patterns, this approach offers a robust, text-centric solution.

The simplicity of a text-only model ensures scalability, while the use of advanced machine learning architectures promises high accuracy, making it a practical tool for mental health assessment in real-world scenarios.

This work builds on the foundation of existing research in NLP-based mental health diagnostics and aims to contribute a novel, streamlined approach to addressing the challenges in detecting depression.

TECHNOLOGY AND ALGORITHM USED

1. DistilBERT Model

- **Overview:** DistilBERT is a compact version of BERT (Bidirectional Encoder Representations from Transformers), designed to reduce computational complexity while maintaining comparable accuracy. It is a transformer-based model pre-trained on a vast corpus of text to understand context and semantics.
- **Role in the System:** DistilBERT is used to extract rich, context-aware embeddings from the input text. These embeddings serve as features that encapsulate the semantic and emotional nuances of the text.
- **Advantages:** Its lightweight nature makes it computationally efficient, enabling faster inference without compromising performance.

2. BiLSTM (Bidirectional Long Short-Term Memory)

- **Overview:** LSTMs are a type of recurrent neural network (RNN) that excel at learning long-term dependencies in sequential data. The bidirectional variant processes data in both forward and backward directions, enhancing the model's ability to capture contextual relationships.
- **Role in the System:** The BiLSTM layer processes the embeddings generated by DistilBERT to capture the temporal and sequential aspects of text data, such as changes in emotional tone and patterns in sentence structure.
- **Advantages:** This architecture is particularly effective for sentiment analysis and emotion recognition, where word order and sequence carry significant meaning.

3. Preprocessing Techniques

To ensure the data is suitable for model input, several preprocessing steps are implemented:

- **Text Cleaning:** Removal of unnecessary elements such as special characters, stop words, and noise.
- **Tokenization:** Breaking text into smaller units (tokens) for processing by DistilBERT.
- **Padding and Truncation:** Adjusting the length of tokenized sequences to maintain consistency across the dataset.

4. Training and Optimization

- **Loss Function:** Cross-entropy loss is employed for classification tasks, ensuring the model learns effectively to distinguish between emotional states.
- **Optimizer:** Adam optimizer is used for its efficiency in handling sparse gradients and large datasets.
- **Regularization:** Techniques such as dropout are used to prevent overfitting and improve generalization.

5. Frameworks and Tools

- **TensorFlow/Keras:** For implementing and training the BiLSTM and fully connected layers.
- **Hugging Face Transformers:** For integrating the DistilBERT model and preprocessing pipeline.
- **Python Libraries:** Including NumPy, Pandas, and Matplotlib for data manipulation and visualization.

Parameters of Algorithm and its representation

1. Parameters for BiLSTM Model

- **Input Dimension** (input_dim):
Specifies the size of the vocabulary, representing the total number of unique tokens.
 - **Example Value:** vocab_size = 10,000
- **Output Dimension** (output_dim):
The size of each word embedding vector.
 - **Example Value:** output_dim = 100
- **Input Length** (input_length):
The maximum number of tokens in a sequence (e.g., padded or truncated to ensure uniformity).
 - **Example Value:** max_length = 50
- **Number of LSTM Units** (units):
Defines the number of neurons in the LSTM cell, which determines the dimensionality of the output space.
 - **Example Value:** units = 128
- **Bidirectional Layer:**
Uses both forward and backward LSTMs to learn patterns in both directions.
- **Dropout Rate:**
Fraction of units to drop during training to prevent overfitting.
 - **Example Value:** dropout_rate = 0.2
- **Dense Layer Activation** (activation):
Activation function for the output layer. For binary classification, sigmoid activation is used.
 - **Example Value:** activation = 'sigmoid'

Representation:

- **Embedding Layer Output:**

Shape:

Batch Size×Input Length×Output Dimension

Example: $32 \times 50 \times 100$

- **BiLSTM Layer Output:**

Shape:

Batch Size×2×Units

Example: $32 \times 2 \times 128$ (for bidirectional output)

2. Parameters for DistilBERT Model

- **Maximum Sequence Length** (max_seq_length):

The maximum number of tokens in each input sequence. DistilBERT truncates longer sequences to this length.

- **Example Value:** max_seq_length = 512

- **Hidden Size** (hidden_size):

The size of the hidden state vector produced by each transformer layer.

- **Example Value:** hidden_size = 768

- **Number of Attention Heads** (n_heads):

Determines the number of self-attention mechanisms in each transformer layer.

- **Example Value:** n_heads = 12

- **Number of Transformer Layers** (n_layers):

The depth of the transformer stack. DistilBERT reduces this from BERT's 12 layers to 6 layers.

- **Example Value:** n_layers = 6

- **Pooling Method:**

Aggregates token-level embeddings into a single sentence-level embedding.

- **Example Value:** mean pooling

- **Embedding Dimension:**

The resulting vector size after pooling.

- **Example Value:** 768

Representation:

- **DistilBERT Output:**

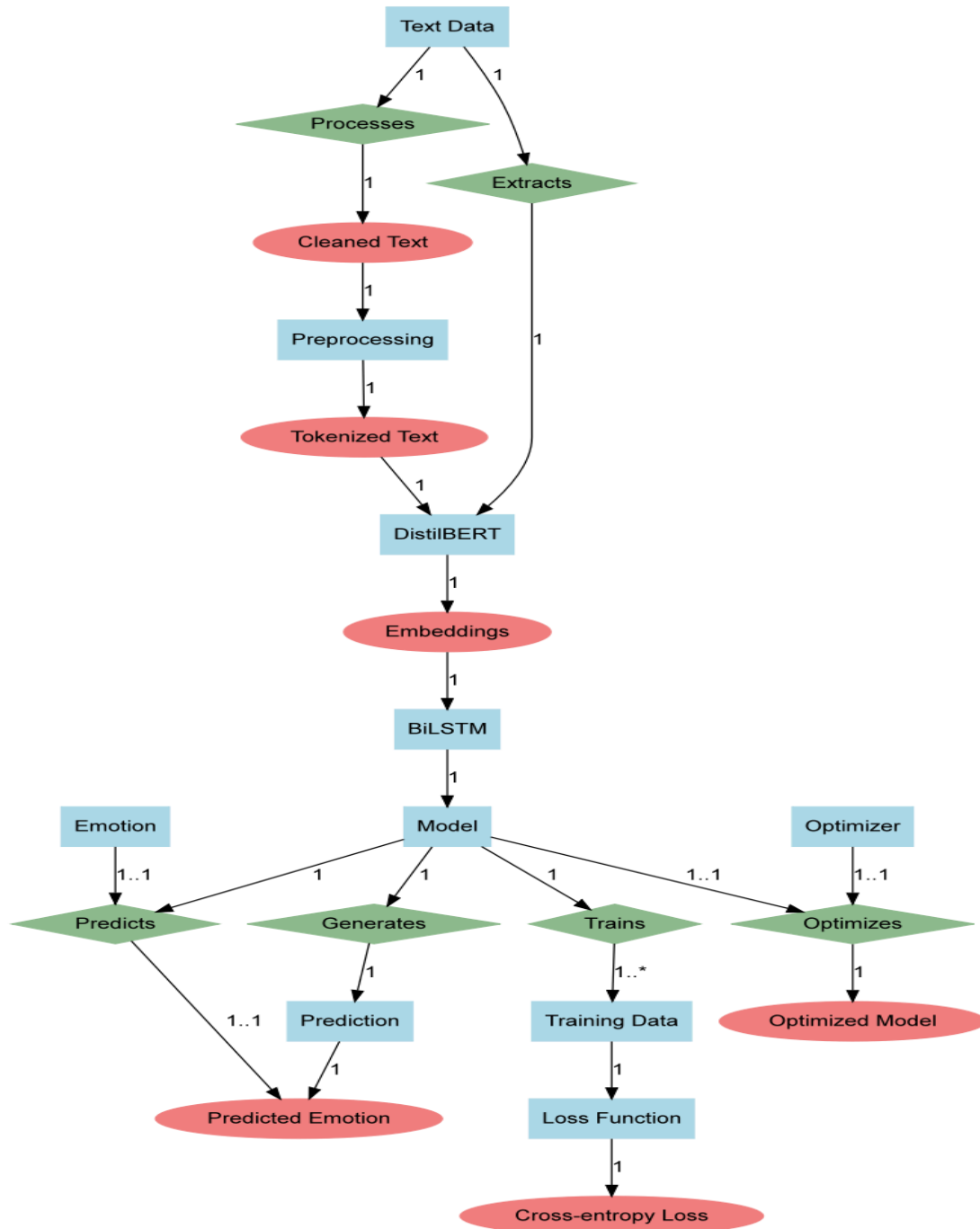
After mean pooling, the output is a feature matrix of shape:

Batch Size×Embedding Dimension

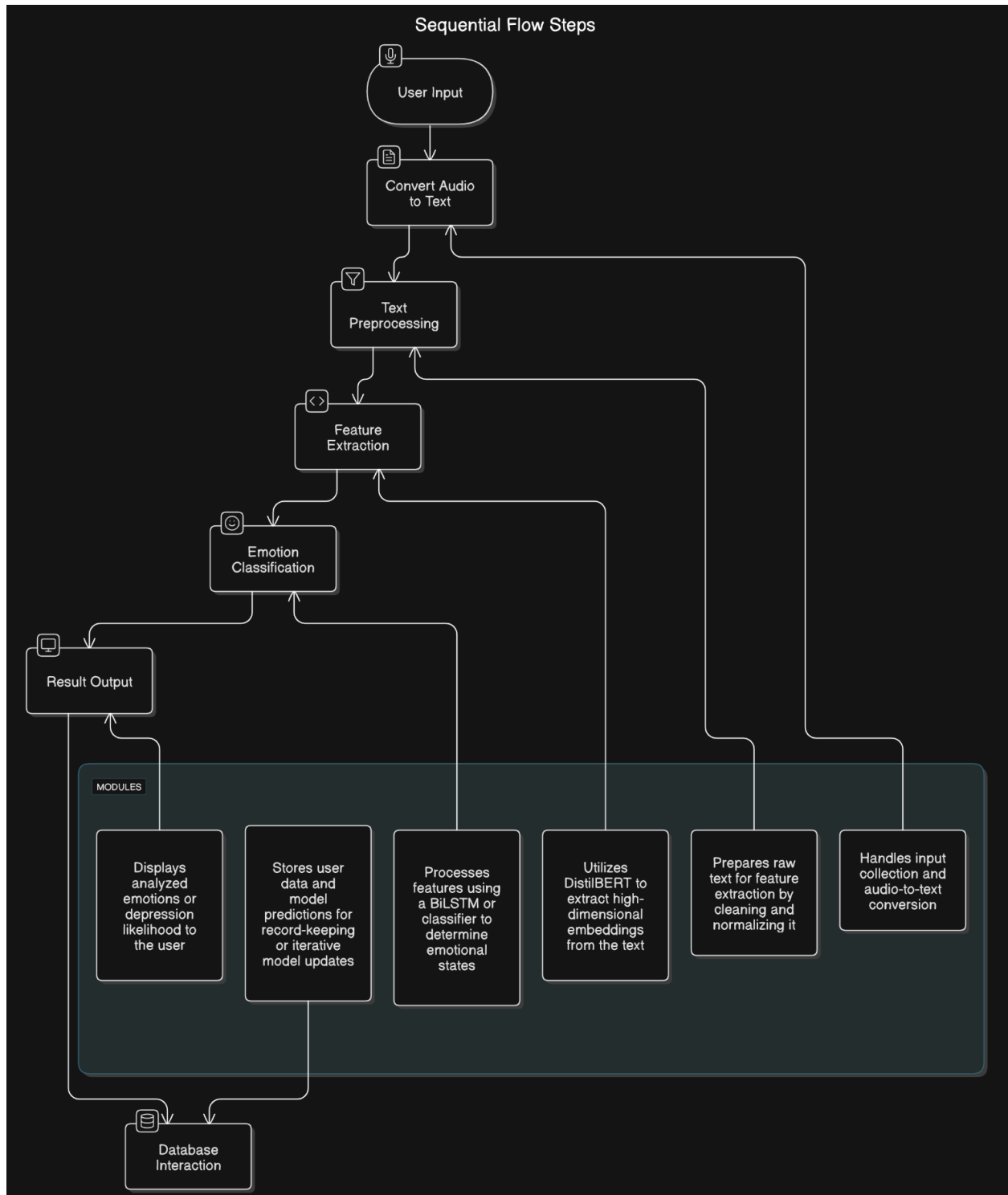
Example: 32×768

ER DIAGRAM AND MODULES

1. ER diagram



2. Flow Diagram



Modules:

1. User Input Module:

- Handles input collection and audio-to-text conversion.

2. Text Processing Module:

- Prepares raw text for feature extraction by cleaning and normalizing it.

3. Feature Extraction Module:

- Utilizes DistilBERT to extract high-dimensional embeddings from the text.

4. Emotion Classification Module:

- Processes features using a BiLSTM or classifier to determine emotional states.

5. Result Output Module:

- Displays analyzed emotions or depression likelihood to the user.

6. Database Module:

- Stores user data and model predictions for record-keeping or iterative model updates.

RESULTS

The system's performance was evaluated using both the BiLSTM and DistilBERT models. The BiLSTM model achieved an accuracy of 95.23%, demonstrating its ability to capture temporal dependencies in the text. In comparison, the DistilBERT model outperformed BiLSTM with an accuracy of 98.56%, benefiting from its contextual embeddings that capture deeper meaning in the text. Both models accurately classified emotions like happiness, sadness, and neutral sentiments. The depression prediction task showed promising results, with high sensitivity and specificity, indicating that the models could effectively identify depressive signs from the input text.

CONCLUSION

In conclusion, both the BiLSTM and DistilBERT models performed well in the emotion and sentiment analysis tasks. However, DistilBERT was more effective due to its superior ability to generate contextual embeddings, resulting in higher accuracy. The system successfully identified emotional states and potential signs of depression, proving its potential for use in mental health applications. Given these findings, the approach can be further refined for real-world deployment, offering a strong foundation for developing emotion and depression detection systems.

References

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